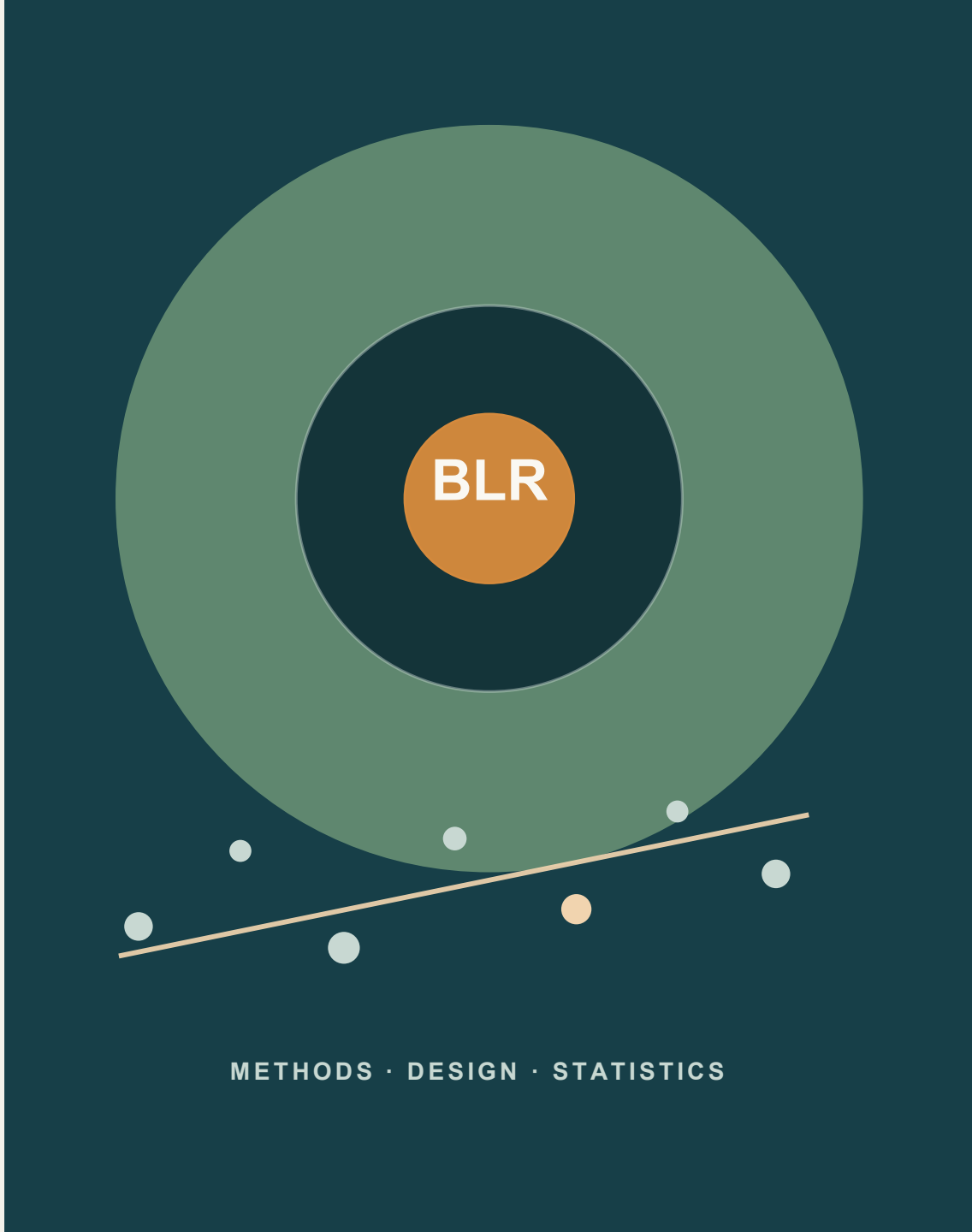
BEYOND MULTIPLE LINEAR REGRESSION

Randomized block designs

Modeling center effects and comparing treatments

Dr Douwe Postmus

UMCG · Department of Epidemiology



Why go beyond multiple linear regression?

Ignoring grouping can violate the usual independent and identically distributed (iid) error assumptions. This may introduce bias in estimated effects and/or lead to incorrectly reported uncertainty.

GROUPED OBSERVATIONS

Examples include patients within hospitals, teeth within patients, and pupils within schools.

MODELS THAT ACCOUNT FOR GROUPING

Linear mixed-effects models combine overall effects with group-specific variation.

THE SCOPE OF THIS COURSE

We study experimental designs, hierarchical data and repeated measurements over time.

Our focus is choosing, fitting and interpreting models that reflect how the data arise.

How the course works

A weekly cycle of preparation, guided practice and independent application.

WHEN	WHAT YOU DO
Before class	Read the selected book chapters to prepare for the session.
During class	Build understanding through lectures and develop data analysis skills in guided R labs.
After class	Apply the readings and in-class material to a new dataset in the weekly assignment.

Assignments and reproducible reporting

Use Quarto to bring your analysis and interpretation together.

ONE REPRODUCIBLE REPORT

A Quarto source file (.qmd) combines written explanations with R code.

Rendering runs the analysis and produces a report containing the results.

Explain your choices and interpret the findings alongside the analysis.

WHAT TO SUBMIT

The Quarto source file (.qmd), including including all R code and annotations.

The rendered HTML report, with the R code visible.

Use the provided assignment template.

Submit both the source and the rendered report so the analysis can be read and reproduced.

Certificate and assessment

COURSE CERTIFICATE

Participate in at least 80% of the in-class meetings.

Submit the weekly data analysis assignments.

IF YOU REQUIRE A GRADE

Your grade is based on the final data analysis assignment.

This applies, for example, to research master students and PhD students seeking Epidemiologist B registration.

PART 1

A multicenter trial: initial analysis





A multicenter rehabilitation trial

DESIGN AND OUTCOME

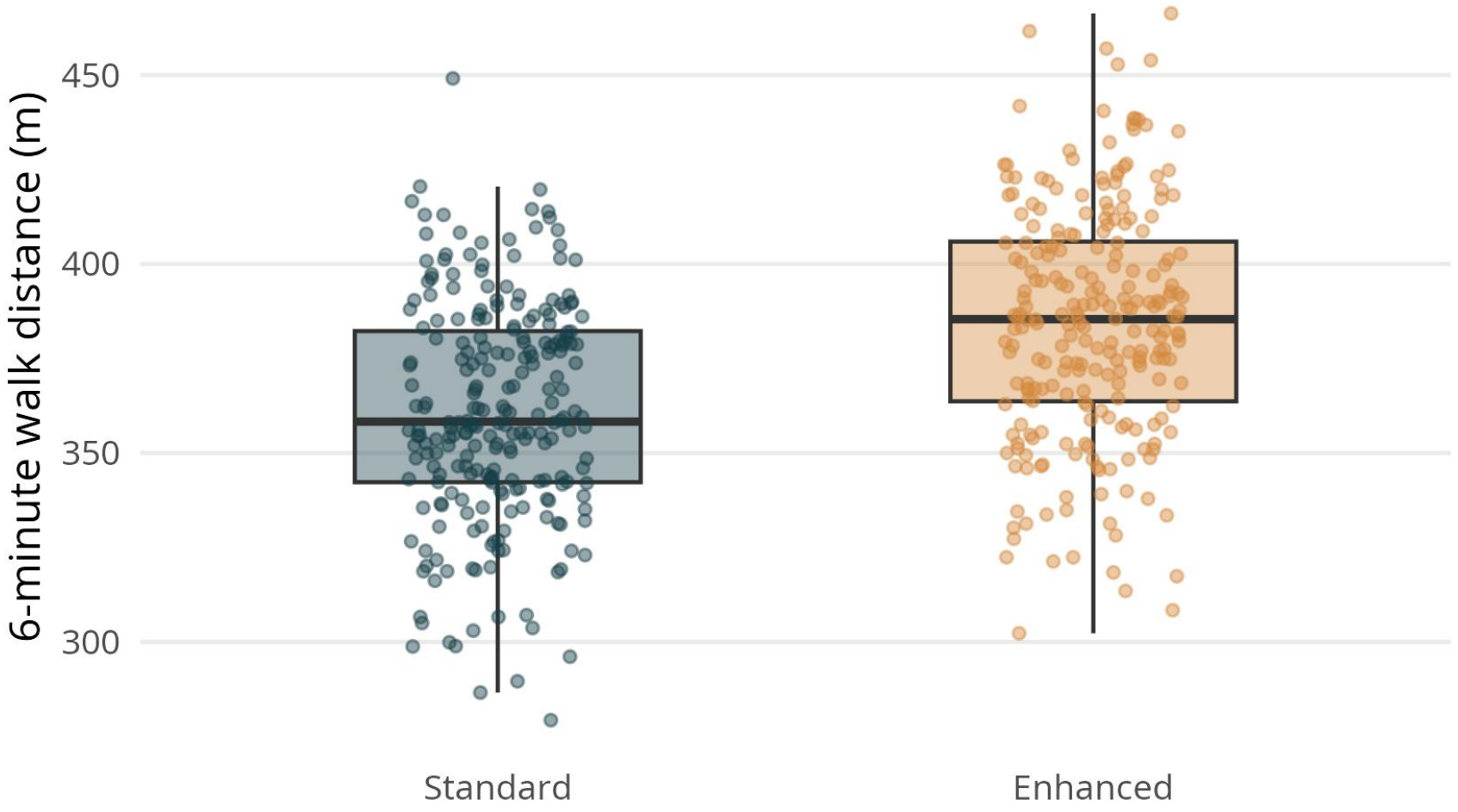
- Simulated trial: 480 patients after knee replacement at 6 centers.
- Randomized block design: centers are the blocks. Within each center, 40 patients are randomized to each treatment.
- Standard care: weekly physiotherapy and home exercises.
- Enhanced care: standard care plus a weekly supervised session.
- Outcome: six-minute walk distance at 12 weeks.

RESEARCH QUESTION

How much do the mean walking distances differ between the programs?



Walking distances pooled across centers



240 patients per program across all 6 centers. Each point represents one patient.

Start with a model for the outcome

For patient i in center j: Y_{ij} is walking distance and T_{ij} indicates the assigned treatment.

SYSTEMATIC PART AND ERROR

$$Y_{ij} = \underbrace{\beta_0 + \beta_1 T_{ij}}_{\text{model-predicted outcome}} + \underbrace{u_{ij}}_{\text{error}}$$

The model predicts an outcome from the patient's treatment.

u_{ij} is the error: the difference between the the observed outcome and that prediction.

INTERPRETING THE PREDICTION

Standard care ($T_{ij} = 0$):

$$\beta_0$$

Enhanced care ($T_{ij} = 1$):

$$\beta_0 + \beta_1$$

β_1 is the difference in model-predicted outcomes between enhanced and standard care.

To complete the statistical model, we need to specify how the errors behave.

Mean, variance and covariance of the errors

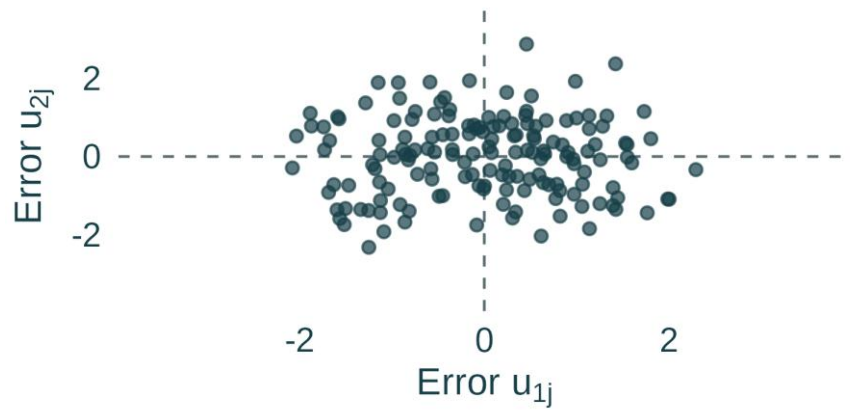
CONCEPT	MEANING FOR THE ERRORS	TREATMENT-ONLY MODEL
Expected value	The average error over many repetitions.	Zero for every patient. $$E[u_{ij}] = 0$$
Variance	How widely a patient's error varies around its mean.	The same for every patient. $$\mathrm{Var}(u_{ij}) = \sigma_u^2$$
Covariance	How two patients' errors vary together around their respective means.	Zero for every pair of different patients. $$\mathrm{Cov}(u_{1j}, u_{2j}) = 0$$

These quantities describe where errors are centered, how much they vary, and how they vary together.

Independence implies zero covariance

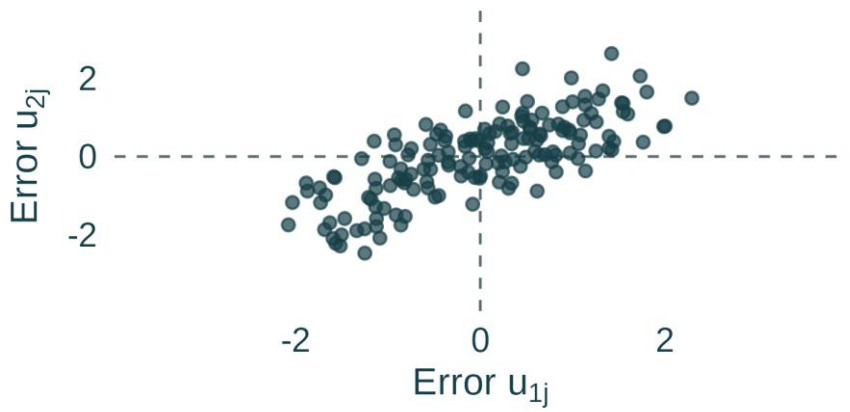
Each point pairs two patients' errors from one repetition. Left: independently generated errors.

INDEPENDENT ERRORS



Knowing one patient's error gives no information about the other's. Their covariance is zero.

DEPENDENT ERRORS: POSITIVE COVARIANCE



The errors tend to move together: above or below their respective means.

Both panels have the same means and variances. The dashed lines mark the expected errors.

Completing the statistical model

TREATMENT-ONLY MODEL

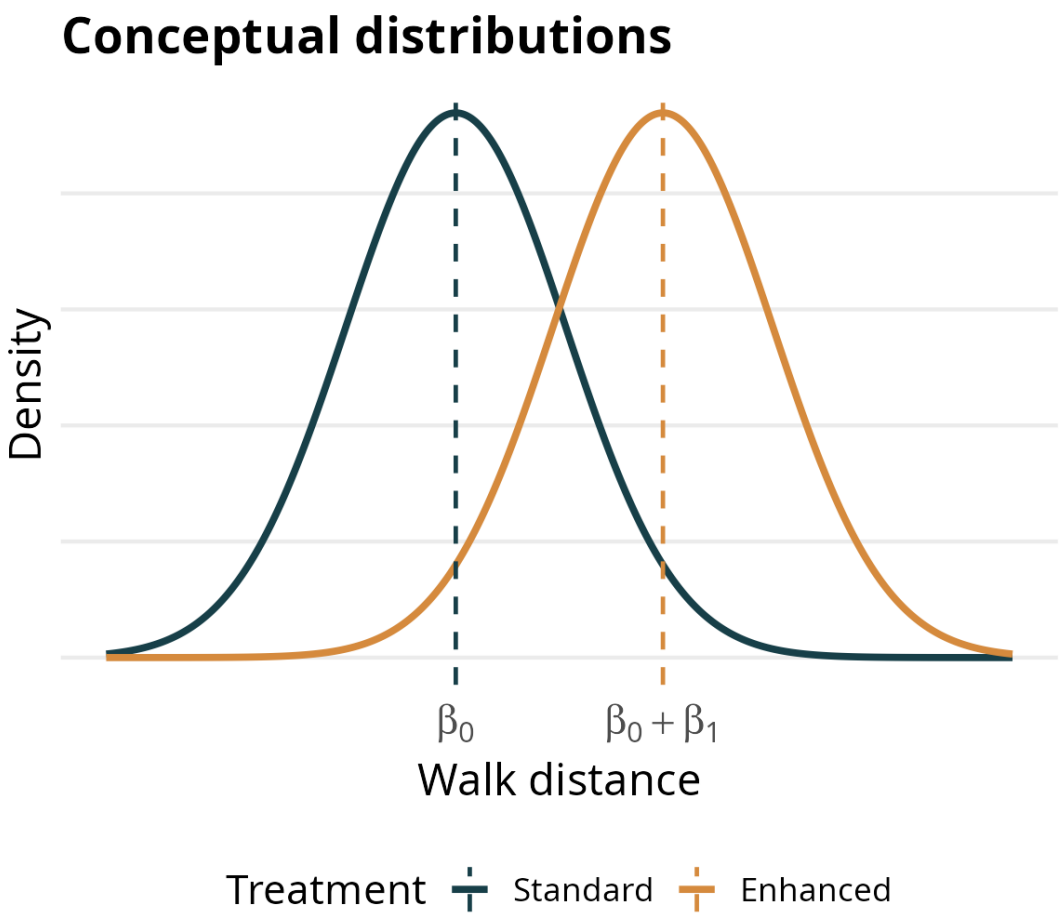
$$Y_{ij} = \beta_0 + \beta_1 T_{ij} + u_{ij}$$

$$u_{ij} \overset{\text{iid}}{\sim} N(0, \sigma_u^2)$$

READING THE NOTATION

- ~ means “is distributed as”.
- N(0, σ_u^2): a Normal distribution with mean zero and variance σ_u^2.
- iid means “independent and identically distributed”: errors are independent across patients and follow the same distribution.

Errors share one distribution. Treatment changes the mean outcome.



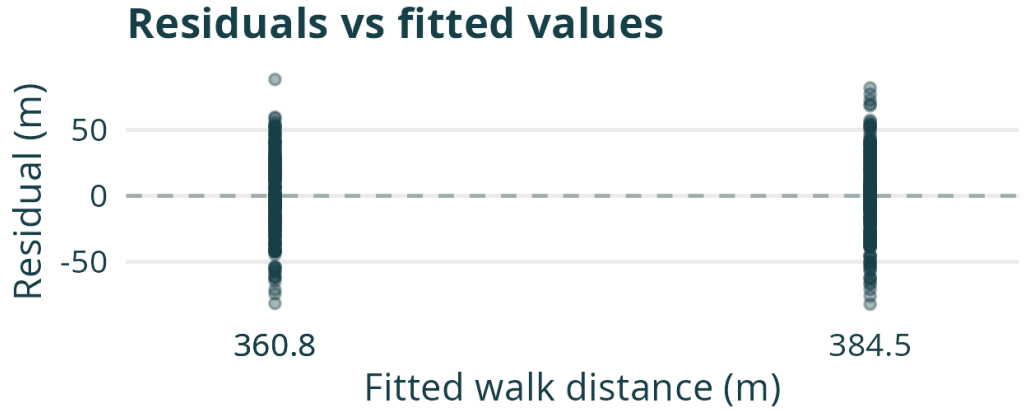
Initial treatment-only results

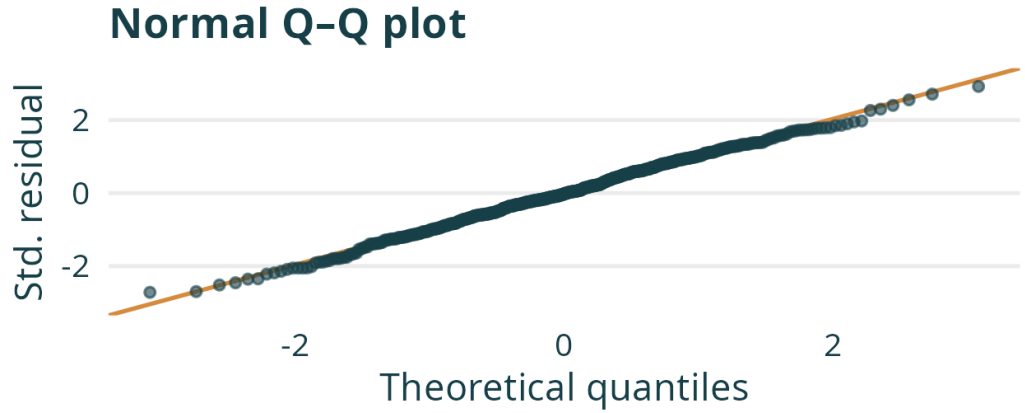
Term	Estimate (m)	SE (m)	95% CI (m)	p-value
Intercept (β_0)	360.8	1.95	357.0 to 364.7	< .001
Enhanced care (β_1)	23.6	2.76	18.2 to 29.1	< .001

Pooling across centers, patients assigned to enhanced care walked 23.6 m farther on average.

Estimated residual variance: 916.4 m².
Corresponding residual SD: 30.3 m.

Checking the treatment-only model





WHAT WE USE THIS PLOT FOR

Assess whether residuals are centered around zero and have similar spread across fitted values.

WHAT WE USE THIS PLOT FOR

Assess whether the residual distribution is approximately Normal.

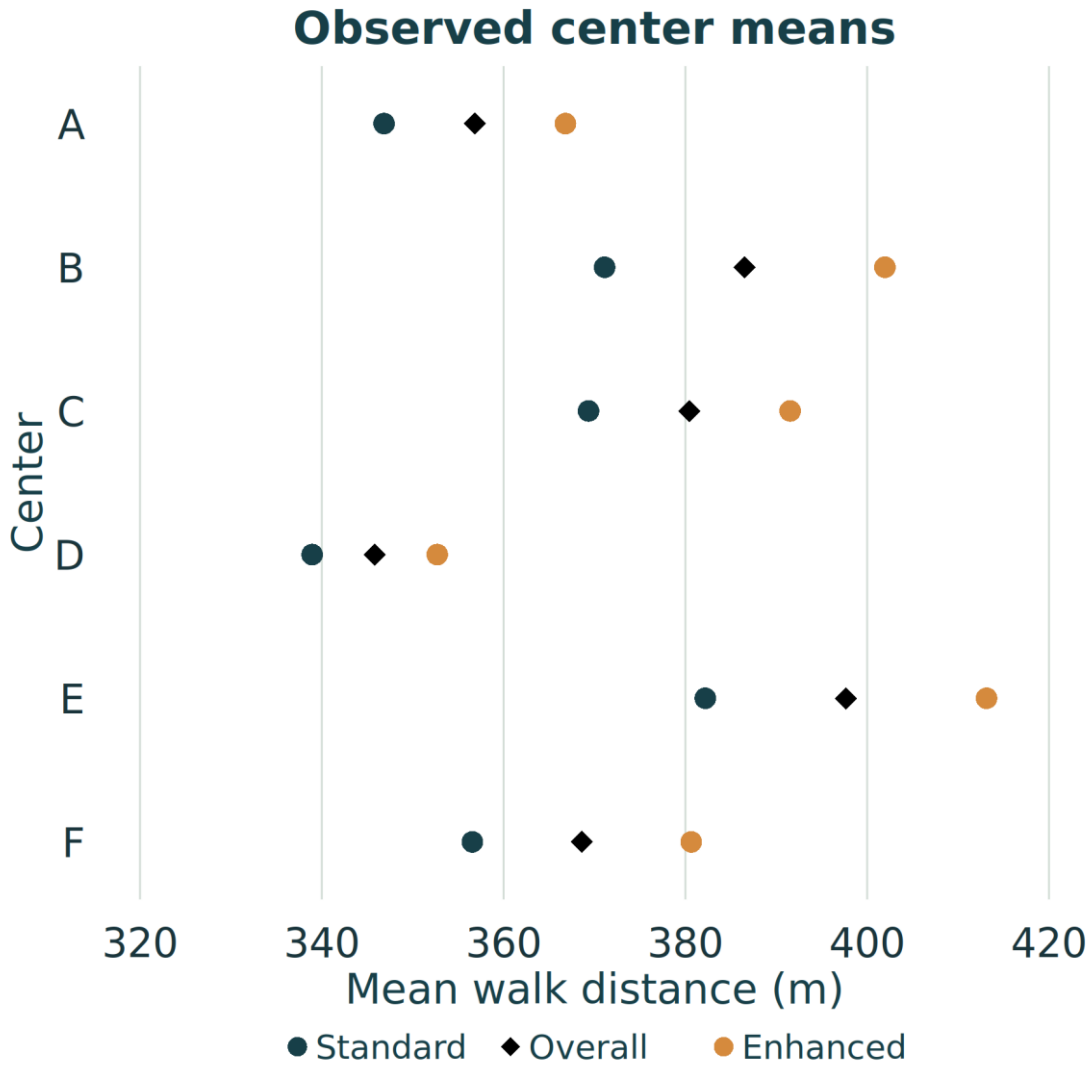
The residual plots do not suggest clear violations of the Normality or equal-variance assumptions.

Does the model reflect the study design?

OUR MODEL AND OUR STUDY

- Our model gives the same predicted outcome to patients receiving the same treatment.
- Patients were recruited from six centers.
- Patient populations and care settings may differ between centers.
- Patients receiving the same treatment may therefore have different expected outcomes across centers.

If centers differ in their expected outcomes, where do those differences enter our treatment-only model?



What happens if we leave center out?

Suppose expected outcomes differ between centers, even under the same treatment.

ACCOUNTING FOR TREATMENT AND CENTER

$$Y_{ij} = \beta_0 + \beta_1 T_{ij} + c_j + \varepsilon_{ij}$$

$$\varepsilon_{ij} \overset{\text{iid}}{\sim} N(0, \sigma^2)$$

c_j: shifts the predicted outcome up or down down for both treatments in center j.

ε_{ij}: the remaining error; independent across patients, Normally distributed with mean zero and common variance.

OUR TREATMENT-ONLY MODEL

$$Y_{ij} = \beta_0 + \beta_1 T_{ij} + u_{ij}$$

Our prediction uses treatment alone. The difference associated with center therefore remains in the error:

$$\boxed{u_{ij} = c_j + \varepsilon_{ij}}$$

Does the original assumption still hold?

$$u_{ij} \overset{\text{iid}}{\sim} N(0, \sigma_u^2) \quad ?$$

If the model on the left is correct, are the errors in our treatment-only model still independent and identically Normally distributed, with mean zero?

Center as a fixed or random effect

What do we imagine changing when we repeat the study?

CENTER AS A FIXED EFFECT

Imagine recruiting new patients in the same centers.

Each center's underlying effect c_j stays the same across repetitions.

We describe the center differences using separate parameters.

CENTER AS A RANDOM EFFECT

Imagine sampling centers from a population of centers, as well as recruiting new patients.

Center effects vary across repetitions according to a distribution.

We describe the center differences using a distribution with an estimated variance.

First, we hold the centers fixed and examine what happens when we repeatedly sample new patients.

PART 2

Modeling center as a fixed effect





Using simulation to study model misspecification

A model is misspecified when it does not adequately describe how the outcomes arise. Simulation lets us investigate the consequences under conditions we control.

1 SPECIFY HOW OUTCOMES ARE GENERATED

Choose a model and its parameter values, so we know what is true in the simulation.

2 GENERATE REPEATED SAMPLES

Repeat the same study many times, drawing new observations each time.

3 EXAMINE A MODEL THAT LEAVES SOMETHING OUT

Check the error assumptions and examine the consequences for estimates and uncertainty.

We start by examining the errors when a treatment-only model leaves out differences between centers.

Step 1: Specify the data-generating model

MODEL AND PARAMETERS

$$Y_{ij} = 100 + 5T_{ij} + c_j + \varepsilon_{ij}$$

$$\varepsilon_{ij} \overset{\text{iid}}{\sim} N(0, 5^2)$$

T_{ij} = 0 for standard care and 1 for enhanced care.

Baseline: 100. Enhanced care adds 5 to the expected outcome.

Individual errors are independent, with mean 0 and SD 5.

EXPECTED OUTCOMES

Center 1: $c_1 = -10$.
Center 2: $c_2 = +10$.

CENTER	STANDARD	ENHANCED
1	90	95
2	110	115

Each center effect shifts both treatment means by the same amount.

This model specifies the outcome distribution for a patient, given their treatment and center.

Step 2: Generate a sample

Sample four patients: one receiving each treatment in each center.

CENTER	TREATMENT	EXPECTED OUTCOME	DRAWN ERROR ε_{ij}	OUTCOME Y_{ij}
1	Standard care	90	-0.7	89.3
1	Enhanced care	95	4.4	99.4
2	Standard care	110	7.4	117.4
2	Enhanced care	115	-6.8	108.2

Draw each error independently from $N(0, 5^2)$, then add it to that patient's expected outcome.

Step 3: Examine the treatment-only errors

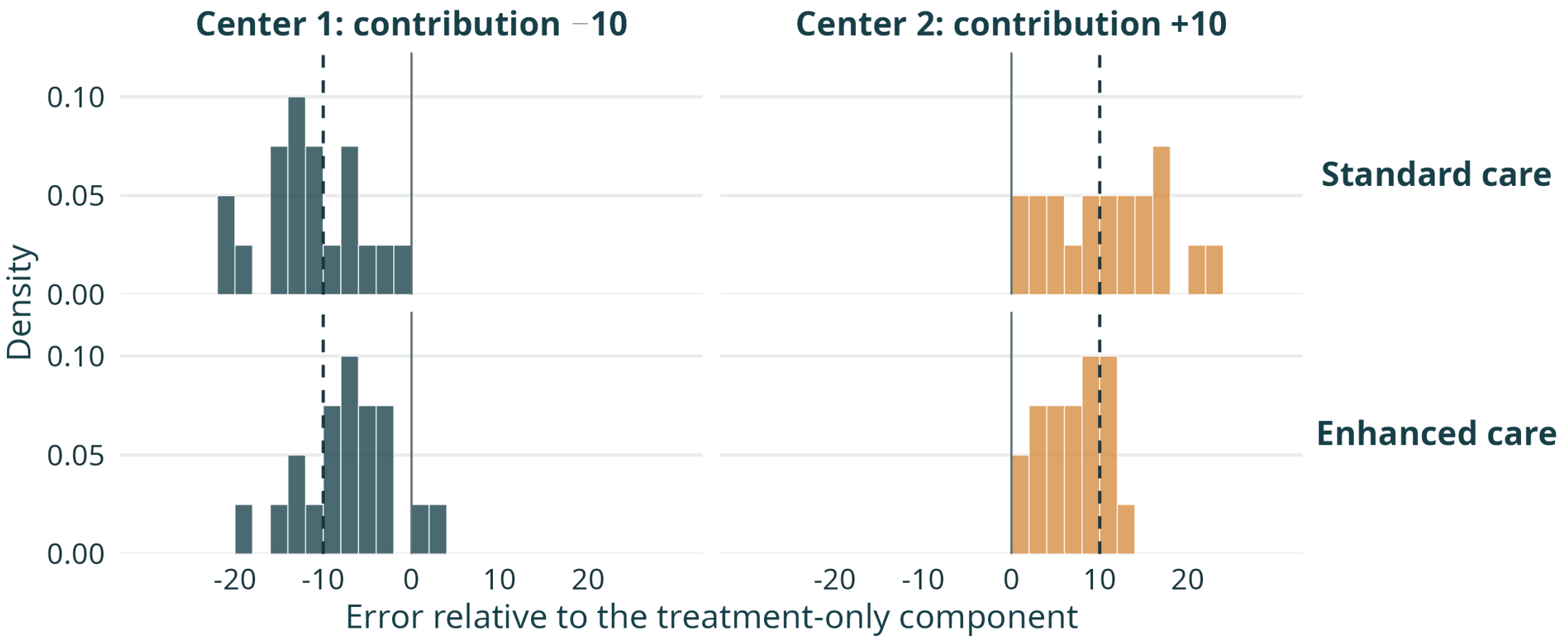
Use the same sample, but leave center out of the prediction.

$$u_{ij} = Y_{ij} - (100 + 5T_{ij}) = c_j + \varepsilon_{ij}$$

CENTER	TREATMENT	OUTCOME Y_{ij}	PREDICTED OUTCOME	ERROR u_{ij}
1	Standard care	89.3	100	-10.7
1	Enhanced care	99.4	105	-5.6
2	Standard care	117.4	100	17.4
2	Enhanced care	108.2	105	3.2

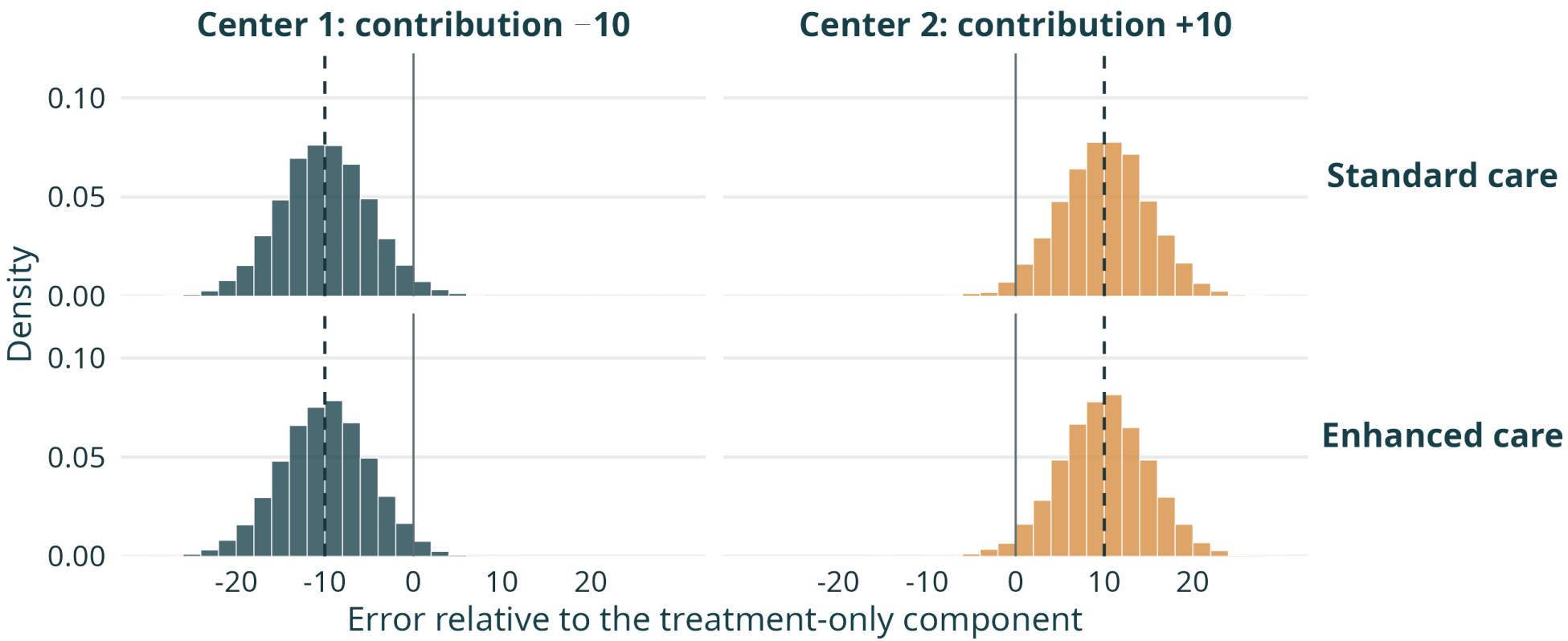
Repeat this process with new patients in the same centers to study the distribution of these errors.

Repeated sampling: 20 studies



Each panel follows one patient position. Dashed line: expected error. Solid line: zero.

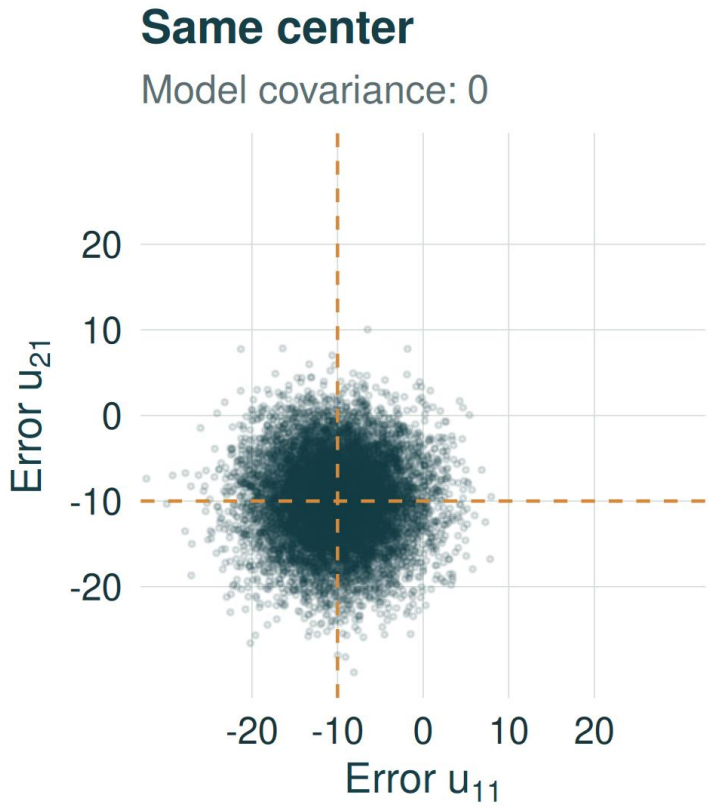
Repeated sampling: 10,000 studies

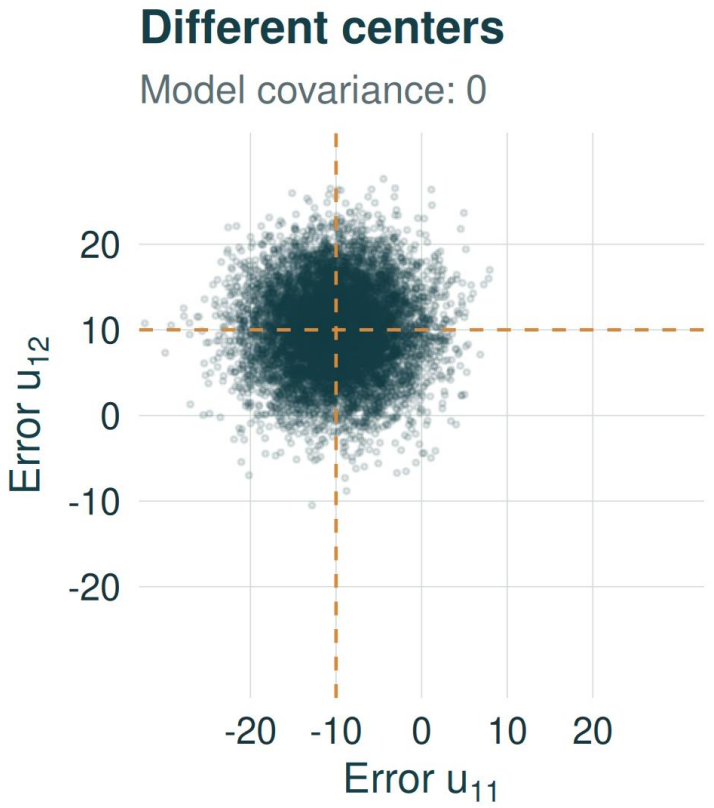


Each panel follows one patient position. Dashed line: expected error. Solid line: zero.

The errors are Normally distributed with equal variance, but their means differ by center: −10 and +10.

Zero covariance within and across centers





Each point pairs errors from one repetition. Dashed lines mark the expected errors.

In this simulation, errors remain independent within and across centers, so their covariance is zero.

Do these errors satisfy our assumptions?

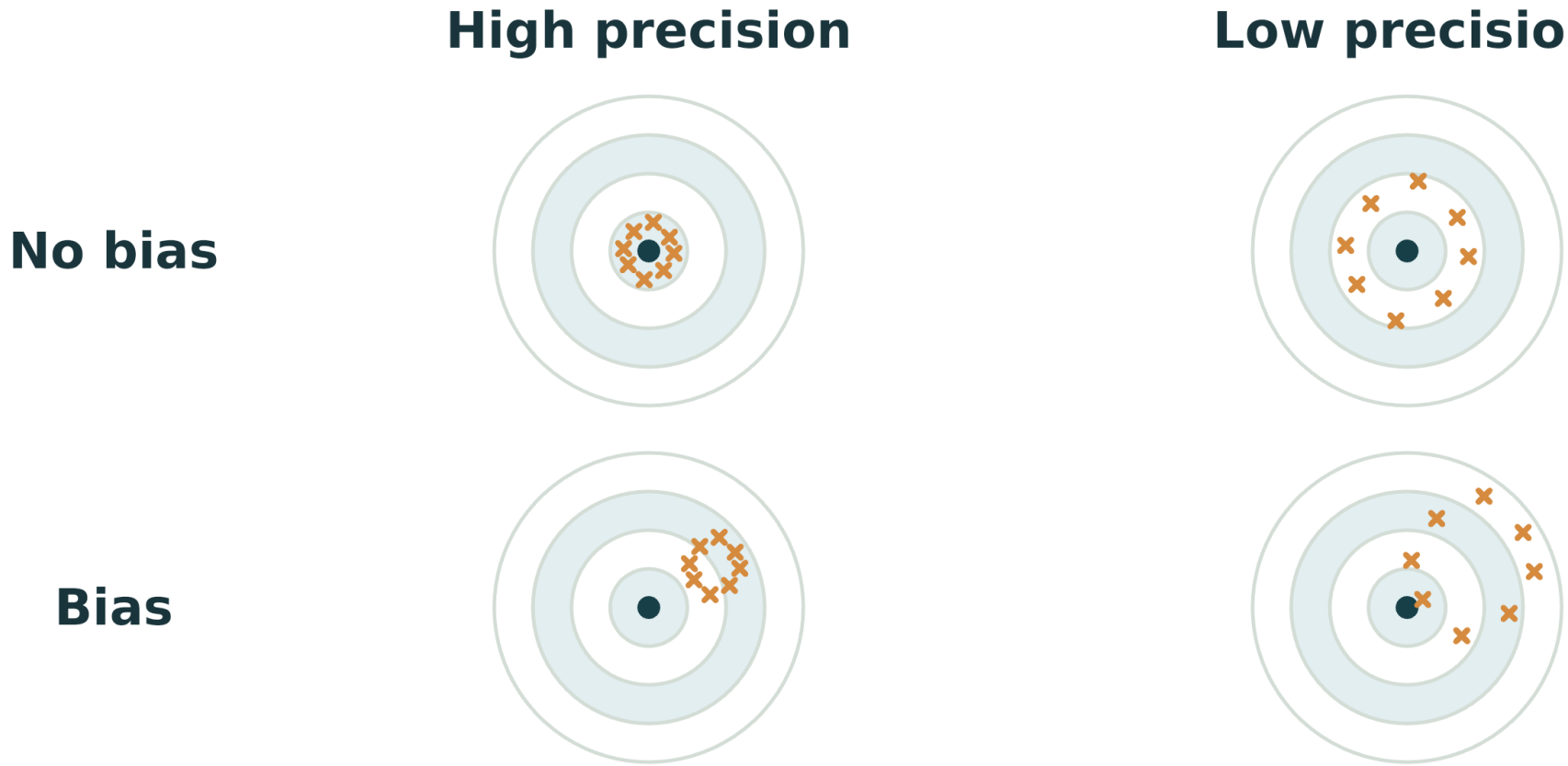
Consider the error distribution for each of the four patient positions.

ASSUMPTION	STATUS	IN THIS SIMULATION
Zero mean	Violated	Mean −10 for both positions in center 1; mean +10 for for both positions in center 2.
Equal variance	Satisfied	Variance 25 (SD 5) for every position.
Independence	Satisfied	Errors vary independently within and across centers, giving zero covariance for both pairs.
Normality	Satisfied	Each error distribution is Normal, centered at its own center’s contribution.

Bias and precision across repeated studies

Bullseye: true treatment effect

Each dart: estimate from one simulated study



Bias: difference between the average estimate and the true treatment effect.
Precision: how tightly the estimates cluster across repeated studies.

What are the consequences of omitting center?

$$E[\widehat{\beta}_1] = \beta_1 + \bar{c}_{\text{enhanced}} - \bar{c}_{\text{standard}}$$

Expected estimate = true treatment effect + difference in average center contributions.

WHERE IS THE ESTIMATE CENTERED?

Balanced allocation within each center gives both groups the same average center contribution.

$$E[\widehat{\beta}_1] = \beta_1$$

The treatment estimate is unbiased.

HOW MUCH DOES THE ESTIMATE VARY?

Center contributions cancel in every repetition with new patients.

Only the individual patient errors make the the estimate vary.

The residual spread also contains center differences, so the usual standard error is too large on average.

Here, balanced allocation gives an unbiased treatment estimate, but its standard error is too large on average.

Time for Lab 1



Lab 1 debrief: accounting for center

Compare the treatment estimates and their uncertainty for the same 480 patients.

TREATMENT ONLY

Treatment difference: 23.6 m

Standard error: 2.76 m

95% confidence interval: 18.2 to 29.1 m

Residual SD: 30.3 m

CENTER AS A FIXED EFFECT

Treatment difference: 23.6 m

Standard error: 2.26 m

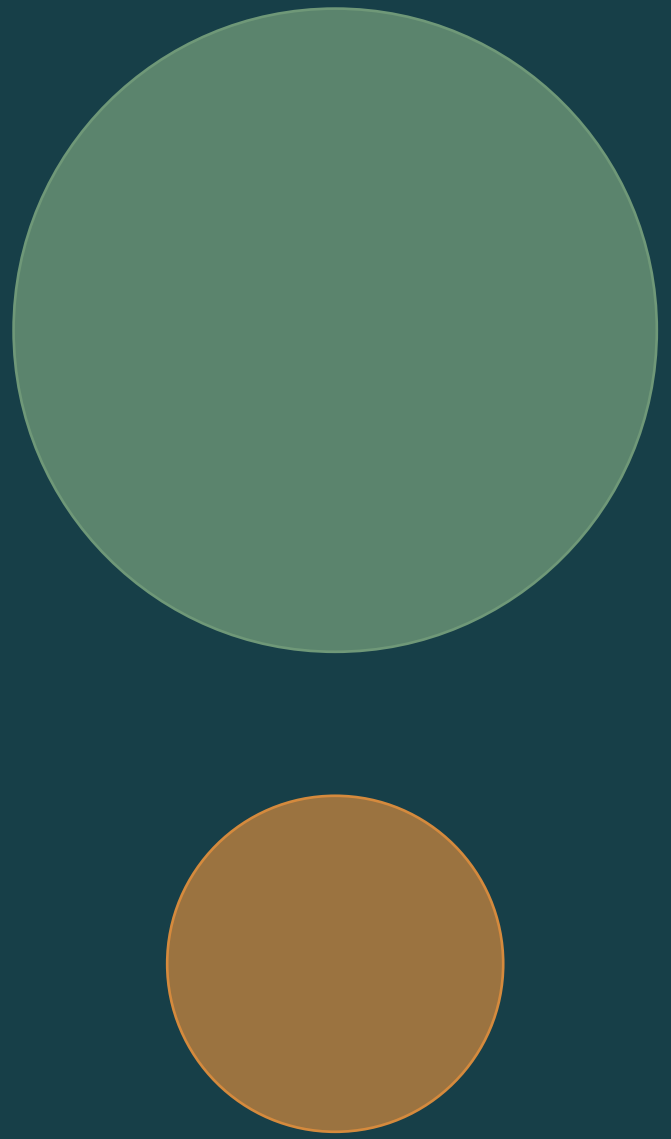
95% confidence interval: 19.2 to 28.1 m

Residual SD: 24.7 m

In this balanced trial, the treatment estimate is unchanged. Accounting for center reduces its standard error.

PART 3

Modeling center as a random effect



Do we want a separate parameter for every center?

The fixed-center model accounts for center differences. Now reconsider how we describe them.

SEPARATE CENTER PARAMETERS

Estimate a mean adjustment for each of the centers in this trial.

This provides a familiar way to adjust the treatment comparison for center.

A MODEL FOR VARIATION ACROSS CENTERS

We may also want to describe how much outcomes vary across a population of centers.

Could we represent the center contributions using a shared distribution?

Next: model center contributions using a distribution, within the same randomized block design.

Step 1: Specify the data-generating model

MODEL AND PARAMETERS

$$Y_{ij} = 100 + 5T_{ij} + c_j + \varepsilon_{ij}$$

$$c_j \overset{\text{iid}}{\sim} N(0, 10^2)$$

$$\varepsilon_{ij} \overset{\text{iid}}{\sim} N(0, 5^2)$$

T_{ij} = 0 for standard care and 1 for enhanced care.

Center effects and individual errors are independent of each other.

WHAT CHANGES ACROSS STUDIES?

Draw new center effects in every repetition: mean 0 and SD 10.

Patients in the same center share the same drawn center effect.

Expected outcomes given c_j:
Standard: 100 + c_j
Enhanced: 105 + c_j

Averaging over centers gives expected outcomes of 100 and 105.

The treatment effect and individual-error distribution stay the same. Center effects now vary across studies.

Step 2: Generate a sample

Draw two center effects, then one patient per treatment in each center (rounded values).

CENTER	TREATMENT	CENTER EFFECT c_j	DRAWN ERROR ε_{ij}	OUTCOME Y_{ij}
1	Standard care	15.8	-0.4	115.4
1	Enhanced care	15.8	3.5	124.4
2	Standard care	-8.2	-1.7	90.2
2	Enhanced care	-8.2	4.0	100.8

Outcome = $100 + 5T_{ij} + c_j + \varepsilon_{ij}$. Each center shares c_j; individual errors are independent.

Step 3: Examine the treatment-only errors

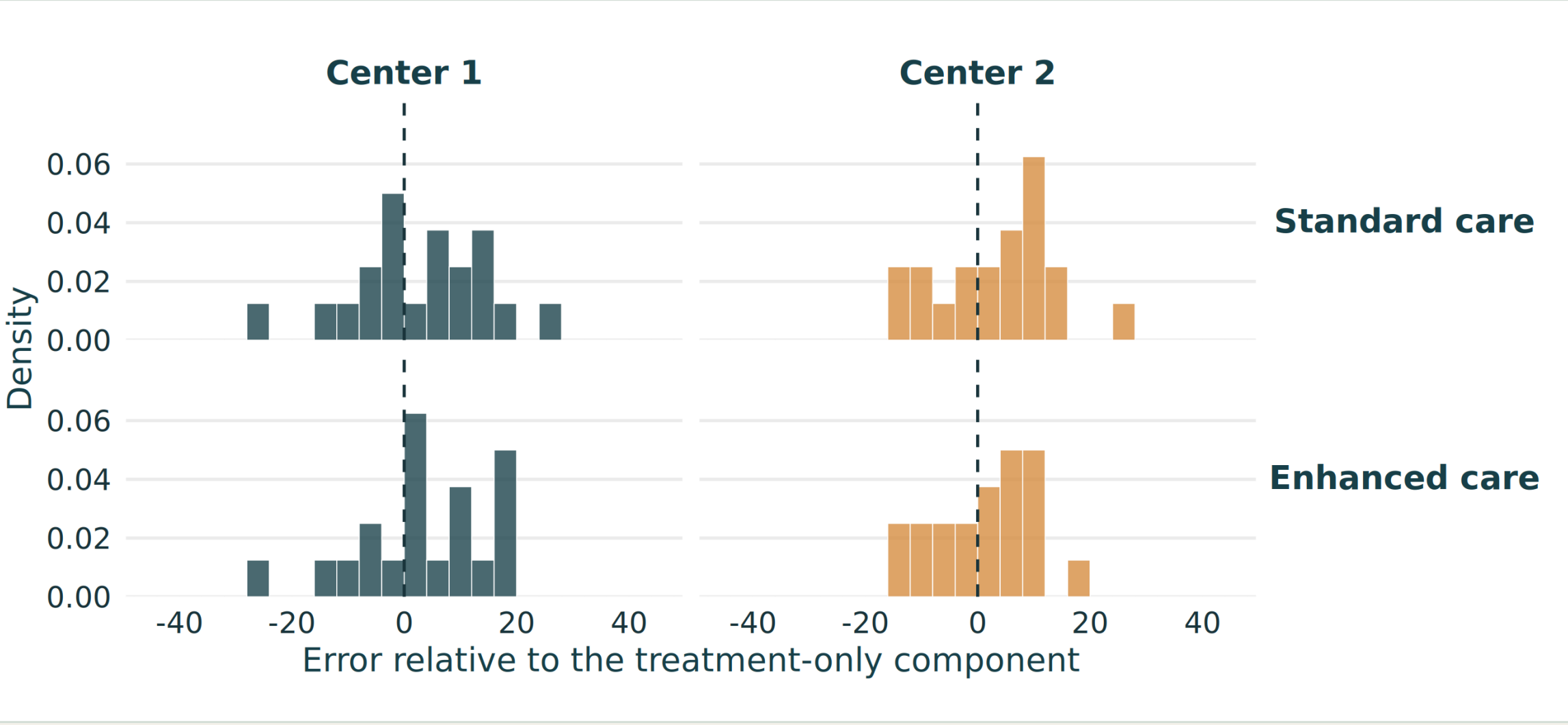
Use the same sample, but leave center out of the prediction.

$$u_{ij} = Y_{ij} - (100 + 5T_{ij}) = c_j + \varepsilon_{ij}$$

CENTER	TREATMENT	OUTCOME Y_{ij}	PREDICTED OUTCOME	ERROR u_{ij}
1	Standard care	115.4	100	15.4
1	Enhanced care	124.4	105	19.4
2	Standard care	90.2	100	-9.8
2	Enhanced care	100.8	105	-4.2

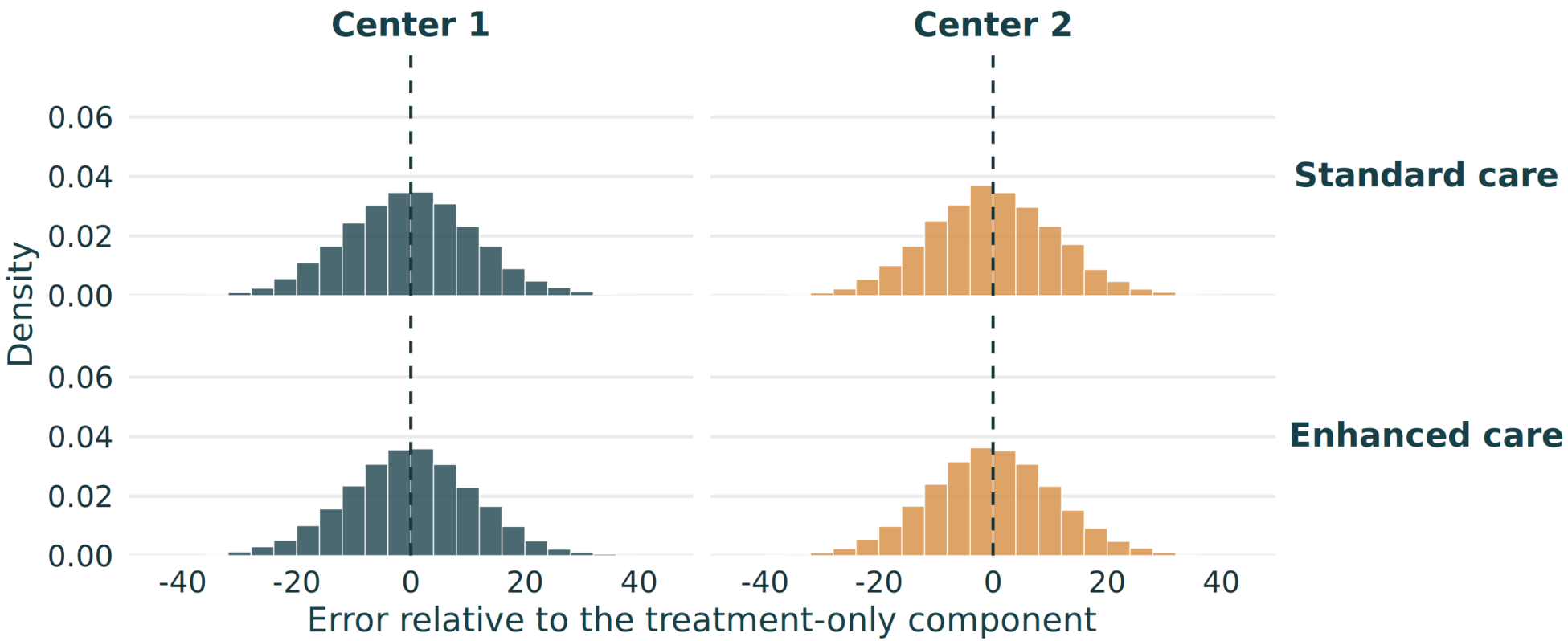
Repeat the study, drawing new center effects and new individual errors each time.

Repeated sampling: 20 studies



Each panel follows one patient position. Dashed line: expected error, now zero in every panel.

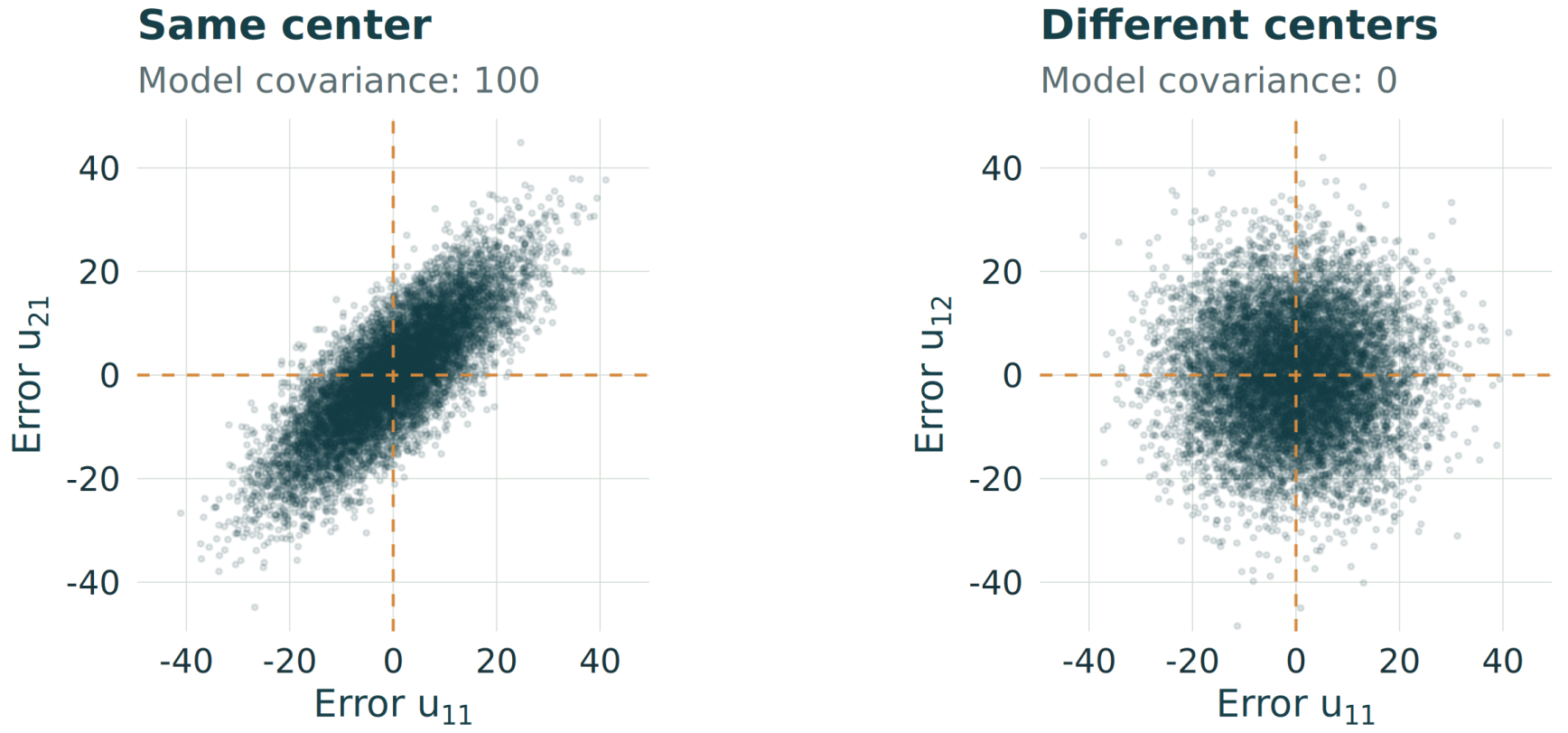
Repeated sampling: 10,000 studies



Each panel follows one patient position. Dashed line: expected error, now zero in every panel.

Averaging over new centers and patients, every error has the same Normal distribution: mean 0 and variance 125.

Errors vary together within the same center



Each point pairs errors from one repetition. Dashed lines mark the expected errors.

Same center: a shared random effect creates covariance 100. Different centers: independent errors, covariance 0.

Do these errors satisfy our assumptions?

Consider repetitions that draw both new center effects and new individual errors.

ASSUMPTION	STATUS	IN THIS SIMULATION
Zero mean	Satisfied	Mean 0 for every position. Average over new centers and patients.
Equal variance	Satisfied	Variance 100 + 25 = 125 for every position.
Independence	Violated	Errors share c_j within a center: covariance 100. Errors from different centers are independent.
Normality	Satisfied	Each error is a sum of independent Normal center and individual contributions.

The treatment-only errors are identically Normally distributed, but are not independent within centers.

What are the consequences of omitting center?

BIAS

Both treatment groups contain the same centers with the same weights.

Center effects cancel in the treatment comparison, even when they vary across studies.

$$E[\widehat{\beta}_1] = \beta_1 = 5$$

The treatment estimate remains unbiased.

PRECISION AND REPORTED UNCERTAINTY

Only individual errors make the treatment estimate vary. Its sampling SD is still 5.

Center differences also contribute to the the treatment-only residual spread.

The usual standard error is too large on average in this balanced design.

Shared center effects create dependent errors, but cancel in the balanced treatment comparison.

Time for Lab 2



Lab 2 debrief: treatment comparison

Enhanced minus standard care: three analyses of the same 480 patients.

MODEL	TREATMENT EFFECT	SE	95% CONFIDENCE INTERVAL
Treatment only	23.6 m	2.76 m	18.2 to 29.1 m
Center fixed	23.6 m	2.26 m	19.2 to 28.1 m
Center random	23.6 m	2.26 m	19.2 to 28.1 m

In this balanced trial, all three treatment estimates agree. Both center-adjusted models give the same, smaller SE.

Lab 2 debrief: variance components and ICC

Interpret the random-intercept model fitted to the rehabilitation data.

ESTIMATED VARIANCE COMPONENTS

Center variance: 366 m²

Individual patient variance: 611 m²

$$\mathrm{ICC} = \frac{\sigma_c^2}{\sigma_c^2 + \sigma^2}$$

Estimated ICC: 0.37

TWO INTERPRETATIONS OF THE ICC

About 37% of the remaining outcome variance is attributable to differences between centers.

For two different patients in one center: estimated outcome correlation = 0.37, after accounting for treatment.

The variance components and ICC describe outcome variation remaining after accounting for treatment.

The End

Questions



